

To: Dr. Fatema ElShahaby

From:

Ziad Ahmed Rabie 23-101292
Farah Sultan 23-101224
Moataz Ashraf 23-101290
Mostafa Ibrahim 24-101476
Hassan Ahmed 23-101330
Mohamed Allam 23-101291

Subject: Proposal & Interim Report – Egyptian Investment Portfolio Analysis (CSE271)

I am submitting this proposal and progress report for the Egyptian Investment Portfolio Analysis project, which addresses two core objectives: building a reproducible data science pipeline for multi-asset financial data, and extracting actionable, data-driven insights for portfolio allocation in emerging market conditions. This submission reflects the foundational work completed to date, including systematic data sourcing, rigorous cleaning, exploratory visualization, and predictive feature engineering.

The following sections provide a structured overview of the project framework and current progress:

Chosen Data Sets -----

EDA Key Findings -----

Results -----

Egyptian Investment Portfolio Analysis

CSE271-Data Science Methodology

Egypt university of Informatics-EUI

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1. Abstract

This project develops a comprehensive data science framework to optimize investment portfolio allocation within Egypt's volatile emerging market. Addressing the challenges of currency devaluation, high inflation, and fragmented financial data, we constructed a reproducible pipeline to process daily historical prices (2015–2026) across equity indices, real estate stocks, gold, and sovereign T-bills. Through rigorous exploratory data analysis and feature engineering, we identified distinct risk-return profiles: gold consistently acted as a macroeconomic hedge with minimal drawdowns, T-bills provided a stable risk-free anchor, while equities exhibited high volatility and fat-tailed return distributions. Machine learning models—including Logistic Regression, XGBoost, Random Forest, and SVR—were deployed to classify market regimes and predict daily returns, achieving robust performance by leveraging technical indicators and volatility-aware segmentation. Monte Carlo optimization and backtesting further validated a diversified portfolio strategy that historically outperformed traditional benchmarks like the EGX30. We conclude that a dynamic, multi-asset approach balancing growth equities, real estate exposure, gold hedging, and T-bill stabilization significantly enhances risk-adjusted returns. Investors should prioritize uncorrelated asset integration and volatility-triggered rebalancing to navigate Egypt's shifting macroeconomic landscape.

2. Introduction

a. Background and Context

Egypt's financial market has experienced significant macroeconomic shifts over the past decade, including currency devaluation, high inflation, and aggressive monetary policy responses. Investors navigating this environment require robust, data-driven tools to evaluate risk and optimize portfolio allocation. This project focuses on constructing an analytical framework for an Egyptian investment portfolio, leveraging historical price data across equities, commodities, fixed income, and real estate stocks. By applying modern data science techniques, we aim to transform raw market data into predictive insights and actionable investment signals.

b. Literature Review

The theoretical foundation of this project rests on Modern Portfolio Theory (MPT), pioneered by Markowitz (1952), which demonstrates that optimal asset allocation relies on maximizing expected return for a given level of risk through diversification and covariance analysis. In emerging markets like Egypt, MPT principles are critically tested by structural volatility, inflationary pressures, and currency depreciation. Empirical reports from the International Monetary Fund (IMF) and the Central Bank of Egypt (CBE) consistently highlight how EGP devaluation amplifies equity risk while simultaneously elevating the attractiveness of hard-asset

hedges such as gold and short-term sovereign debt (IMF, 2023; CBE, 2024). These macroeconomic dynamics necessitate portfolio frameworks that explicitly account for cross-asset correlations and regime shifts.

Recent quantitative finance literature emphasizes the limitations of traditional linear econometric models in capturing non-linear market dynamics and fat-tailed return distributions. Consequently, machine learning techniques—including ensemble methods (e.g., Random Forest, XGBoost) and kernel-based approaches (e.g., SVM, SVR)—have gained traction for regime classification and return forecasting, frequently outperforming conventional benchmarks in volatile environments (Gu, Kelly, & Xiu, 2020). However, few studies apply these methodologies to a comprehensive, multi-asset Egyptian portfolio framework that integrates real estate equities, commodities, and fixed income. This project bridges that gap by combining rigorous data engineering, advanced ML modeling, and interactive visualization to deliver empirically grounded portfolio strategies tailored to Egypt’s unique macroeconomic conditions.

c. Project Objectives

- Build a reproducible data science pipeline for multi-asset financial data
- Perform rigorous EDA to understand risk profiles across asset classes
- Engineer predictive features for ML modeling
- Extract actionable, data-driven insights for portfolio allocation

3. Materials

a. Data Description

i. Data Source & Collection Protocol

Historical financial data for this project were gathered from three distinct sources to ensure robustness and cross-validation:

Source	Role in Project	Rationale
Egyptian Exchange (EGX) Official Portal	Reference/Validation	Primary official source for Egyptian market data; used to verify index composition and trading calendar accuracy
investing.com	Primary Data Source	Selected for final dataset due to consistent formatting, complete historical coverage (2015–2026), and minimal preprocessing requirements

Yahoo Finance	Reference/Validation	Used for cross-checking price levels and return calculations, particularly for internationally traded instruments (e.g., Gold/EGP) + Benchmark (S&P500)
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ii. Source Selection Justification

After initial exploration of all three platforms, investing.com was designated as the primary source for the analytical dataset based on the following criteria:

Criterion	investing.com	EGX Official	Yahoo Finance
Formatting Consistency	Standardized CSV export	Mixed formats (PDF/HTML)	Standardized, but occasional symbol mismatches
Column Structure Clarity	Explicit labels (Price, Vol., Change%)	Arabic/English mix; requires translation	Clear labels, but EGX-specific symbols sometimes missing
Preprocessing Effort Required	Lowest	High (scraping + parsing)	Medium (symbol mapping + alignment)
Update Frequency	Daily	Daily (post-market)	Daily (real-time)

Consequently, all final modeling and analysis were performed on the investing.com-derived dataset. Data from EGX Official and Yahoo Finance were retained solely for **validation purposes**: spot-checking price levels, confirming return calculations, and identifying potential outliers or reporting discrepancies.

iii. Chosen Data Sets

For the primary dataset (investing.com), extraction was performed with the following specifications:

- Start Date: January 1, 2015
- End Date: May 1, 2026
- Frequency: Daily (trading days only, aligned with EGX calendar)
- Currency: All values denominated in Egyptian Pounds (EGP)
- Reproducibility: Direct URL

Asset	Type	Description and URL
EGX30	Stocks	Benchmark equity index tracking the top 30 most liquid large-cap companies on the Egyptian Exchange URL: https://www.investing.com/indices/egx-100-historical-data
EGX100	Stocks	Broader equity index covering the top 100 companies by market cap and liquidity on the EGX. URL: https://www.investing.com/indices/egx-100-historical-data
Gold (EGP/g)	Commodity	Gold price denominated in Egyptian Pounds per gram, derived from global gold (USD) × USD/EGP exchange rate URL: https://www.investing.com/currencies/gau-egp
1-Year T-Bills	Fixed Income	Short-term sovereign debt instruments issued by the Central Bank of Egypt; maturity ~364 days; yield quoted as annual %. URL: https://www.investing.com/rates-bonds/egypt-1-year-bond-yield-historical-data
EGXREF	Physical Real Estate Fund	Physical real estate fund URL: https://www.investing.com/equities/egyptians-real-estate-fund

iv. Raw Data Parameters

For all data sets (EGX30, EGX100, T-Bills, Gold (EGP/g), EGREF) before **the cleaning process**).

Column	Data Type	Description
Date	object (string)	Trading date in M/D/YYYY format (e.g., "4/30/2026", "1/1/2015"); no time component; may contain leading/trailing whitespace
Price	object (string)	Closing price with comma thousands separators (e.g., "51,760.97", "1,234.50"); may include currency symbol in some exports
Open	object (string)	Opening price with comma formatting (e.g., "51,500.00"); same structure as Price
High	object (string)	Daily high price with comma formatting (e.g., "52,100.25")

Low	object (string)	Daily low price with comma formatting (e.g., "51,200.80")
Vol.	object (string)	Trading volume with 'M' suffix for millions (e.g., "637.54M"); may be empty string "" for missing values (e.g., Gold, or isolated gaps like EGX100 on 4/16/2020)
Change%	object (string)	Daily percentage change with sign and '%' symbol (e.g., "-1.19%", "+0.45%", "0.00%"); zero values may appear as "0.00%" or "0%"

The following subsections **Methods** describe the procedures undertaken for data preprocessing, exploratory data analysis, feature engineering, model development, and result visualization.

4. Methods

a. Data cleaning & Preprocessing

Raw CSV files exported from investing.com required systematic preprocessing to resolve formatting inconsistencies, standardize temporal structure, and ensure compatibility with time-series modeling. The pipeline was implemented in Python using pandas and is fully reproducible.

i. Standardization Pipeline Steps

Each asset file passes through an 11-step deterministic workflow:

Step	Action	Implementation Detail	Output / Effect
1	Load Raw File	Read CSV from raw_data/ using config.py mappings	Source file remains unchanged
2	Normalize Headers	Lowercase names, trim whitespace, standardize labels (Vol. → vol , Change% → change_pct)	Consistent column naming
3	Parse Dates	Convert M/D/YYYY strings to ISO datetime , then set as index	Time-aware DataFrame index
4	Sort Chronologically	Sort index ascending (oldest to newest)	Time-series compatibility
5	Filter Date Range	Restrict to 2015-01-01 through 2026-05-01	Focused analysis period
6	Remove Duplicates	Drop duplicate timestamps (keep='first')	Prevents artificial data inflation

7	Clean Price Columns	Strip commas from price, open, high, low; convert to float	Numeric precision for modeling
8	Format Volume	Remove M suffix from vol; convert to float; empty cells → NaN	Volume stored in millions (EGP)
9	Standardize Returns	Strip % from change_pct ; convert to signed float	Ready for return calculations
10	Audit Missing Values	Log NaN counts per column	Transparency for downstream handling
11	Export Cleaned Data	Save to cleaned_data/{asset}_clean.csv	Model-ready output files

Note: *T-Bills follow the same 11-step flow but retain only date and price columns, as yield % serves directly as return and risk-free rate.*

ii. Error Handling & Data Integrity

- **pd.to_numeric(..., errors="coerce")** ensures malformed strings safely convert to NaN rather than triggering pipeline failures.
- Raw files in raw_data/ remain immutable; all transformations are applied in-memory.
- The pipeline logs row counts, dropped duplicates, and missing value distributions for full auditability.

b. Exploratory data analysis (EDA) & Feature Engineering

We present the full workflow using EGX30 as a representative example. The identical pipeline was applied to **EGX100**, **TMG**, **SODIC**, **Palm Hills**, **Gold**, and **T-Bills**. Asset-specific results are summarized in [Section 4.b.iv](#)

Now, we start with steps taken to tackle this part:

1. Step 1: Load & Inspect Cleaned Data

We begin by loading the cleaned EGX30 dataset (egx30_clean.csv) and performing a quick structural audit to confirm readiness for analysis.

Action	Purpose	Output Verified
Load CSV with date as index	Ensure time-series compatibility	Datetime Index, sorted ascending

Print date range & observation count	Confirm temporal coverage	2015-01-04 to 2026-04-30 (~2,754 trading days)
List columns	Verify feature availability	price, open, high, low, vol, change_pct
Check for missing values	Validate cleaning pipeline effectiveness	0 missing values in core columns

```
<class 'pandas.DataFrame'>
DatetimeIndex: 2754 entries, 2015-01-04 to 2026-04-30
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   price       2754 non-null   float64
1   open        2754 non-null   float64
2   high        2754 non-null   float64
3   low         2754 non-null   float64
4   vol         2743 non-null   float64
5   change_pct  2754 non-null   float64
dtypes: float64(6)
memory usage: 150.6 KB
```

Figure 1: Data inspection 1

Out[94]:

	price	open	high	low	vol	change_pct
date						
2015-01-04	8942.65	8926.26	8942.65	8880.21	62.86	0.18
2015-01-05	9011.07	8941.72	9033.07	8931.80	86.76	0.77
2015-01-06	8798.94	8971.93	8971.93	8754.87	105.19	-2.35
2015-01-08	8946.34	8819.61	8954.29	8819.61	82.04	1.68
2015-01-11	8909.75	8945.69	8957.59	8909.75	45.71	-0.41

Figure 2: Data inspection 2

2. Step 2: Core Visualizations (The 7-Plot Standard)

For each asset, we generated seven standardized plots. Below are the EGX30 examples:

Plot	Purpose	Key Insight (EGX30 Example)	So What?
1. Cumulative Returns + Volume	Show long-term growth trajectory & liquidity	EGX30 grew ~3x from 2015–2026, with sharp drawdowns in 2016 (devaluation) and 2020 (pandemic). Volume spikes precede major moves.	Confirms EGX30 is a high-beta equity index: high reward potential but requires timing/risk management.
2. Return Distribution (Histogram + KDE)	Check for normality, fat tails, skew	Returns are leptokurtic (fat-tailed): extreme moves ($>\pm 2\sigma$) occur 3× more often than Gaussian prediction.	Standard risk models (VaR, Sharpe) will underestimate tail risk. We use robust models (SVR, tree ensembles).
3. Target Class Balance	Assess classification difficulty	Label distribution: 65.9% Down/Flat (0), 34.1% Up (1) mild imbalance.	Accuracy alone is misleading; we prioritize F1-Score and ROC-AUC for model evaluation.
4. Feature Separability (Violin Plots)	Test if features distinguish up/down days	RSI, MACD, and Bollinger features show substantial overlap between labels — no single "magic" indicator.	Justifies ensemble ML: models must learn complex feature interactions, not simple rules.
5. Feature Correlation Matrix	Detect multicollinearity & linear predictability	High correlation between momentum features (e.g., $RSI \leftrightarrow bb_pb: 0.87$). Near-zero correlation between features and label (max	It's noted to the modeling team to drop one of the high correlated features in simple models. Weak correlation with label proves it's hard to predicted the financial market
6. Rolling Volatility (30-day, annualized)	Identify volatility clustering & regime shifts	Volatility clusters: calm periods ($\sigma \approx 15\%$) alternate with stress spikes ($\sigma > 40\%$ in 2020).	Supports regime-aware modeling: we segment data by volatility state before training.
7. Historical Drawdowns	Quantify peak-to-trough risk	Maximum drawdown: ~60% (2020). Recovery took ~18 months.	Critical input for portfolio optimization: EGX30 requires diversification with lo

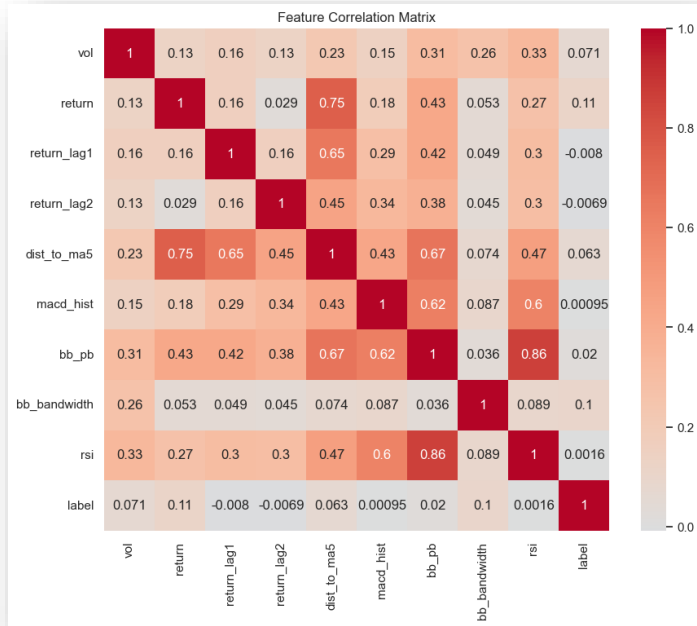


Figure 3: EGX30 Features heats map

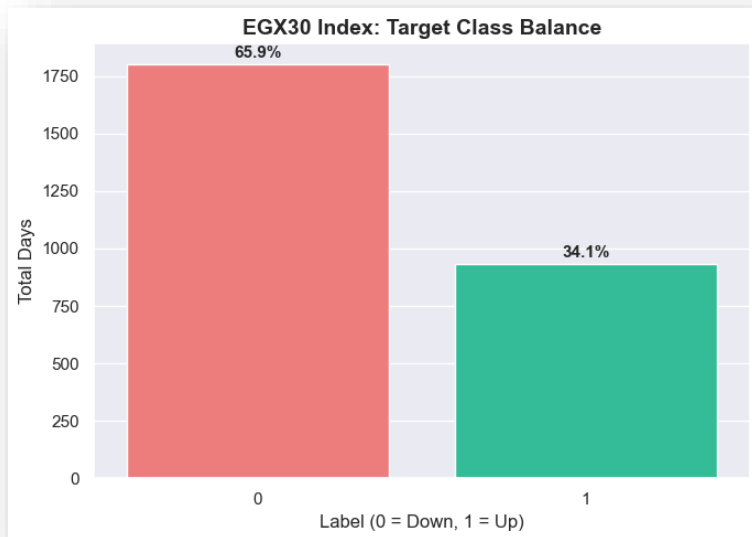


Figure 4: EGX30: Target Class Balance



Figure 5: EGX30, Cumulative Returns

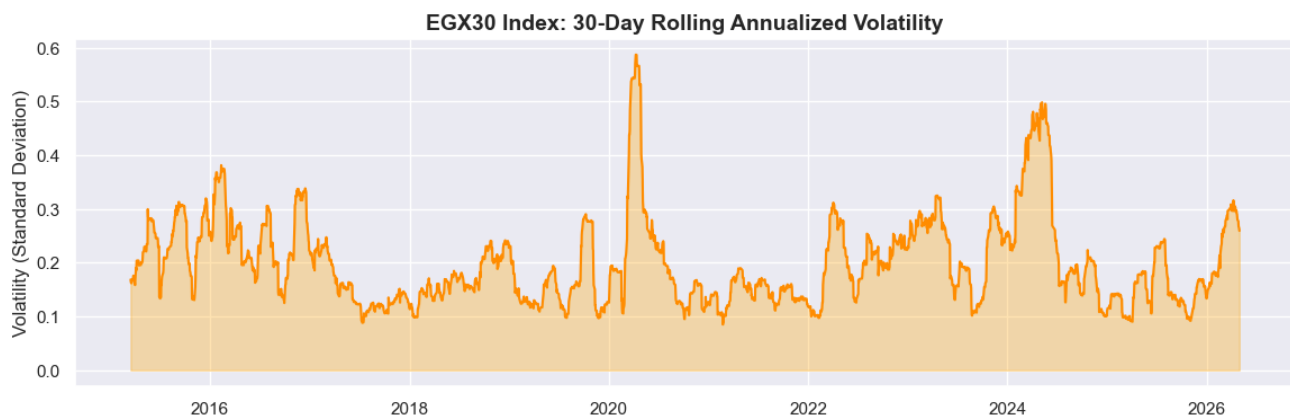


Figure 6: EGX30, Rolling volatility



Figure 7:EGX30, Historical Drawdowns

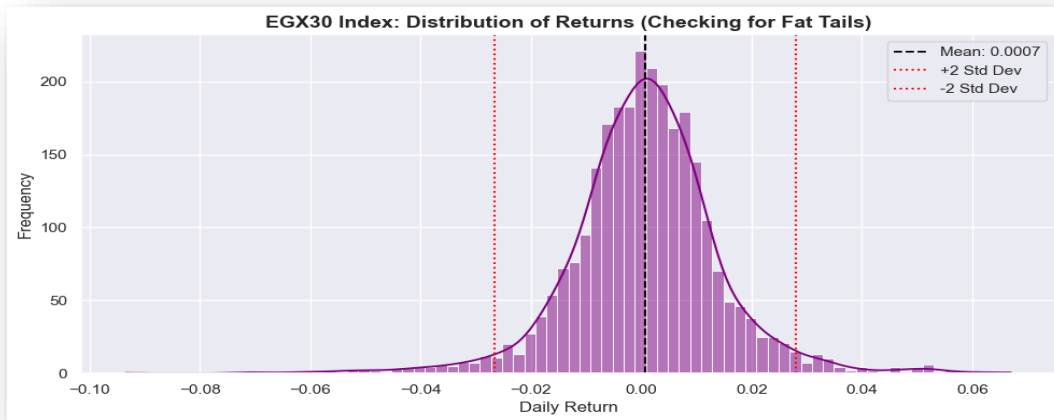


Figure 8:EGX30, Return Distribution

c. Models Chosen

After several trials and testing processes, we found that those models gave the optimal portfolio in the Egyptian market.

i. Classification models

1. Logistic Regression

Purpose: Serves as an interpretable baseline classifier that estimates market regime probabilities using a SoftMax function.

Configuration: Trained on the reduced “lr” feature set using `class_weight="balanced"` with a maximum of 1,000 iterations.

Performance: Achieved accuracy of **56.9%** on both EGX30 and Gold. The weighted F1-score reached **56.7%** for EGX30 and **52.6%** for Gold, with an average weighted F1-score of approximately **54.6%**.

2. SVM (Support Vector Machines)

Purpose: Captures non-linear market regime boundaries by projecting features into a higher-dimensional space.

Configuration: Implemented using an RBF kernel with `C=1.2`, `class_weight="balanced"`, and `probability=True` for calibrated probability outputs.

Output: SVM was preferred over Decision Trees during final selection because Decision Trees achieved only approximately **48.1%** mean accuracy across assets, compared to stronger and more stable performance from Logistic Regression and XGBoost.

3. XGBoost

Purpose: Acts as the primary high-capacity classifier by sequentially correcting prediction errors through gradient-boosted decision trees.

Configuration: Trained with `n_estimators=200`, `learning_rate=0.05`, `max_depth=5`, `subsample=0.8`, and `colsample_bytree=0.8`.

Performance: Achieved the highest classification performance among the tested classifiers, with accuracies of **60.7%** for EGX30 and **55.9%** for Gold. The weighted F1-scores reached **54.0%** and **47.5%**, respectively, resulting in an average accuracy of approximately **58.3%**.

i. Regression models

4. Linear Regression

Purpose: Provides a simple and interpretable baseline by minimizing residual sum of squares.

Configuration: Trained on the reduced “lr” feature set with standard scaling applied before training.

Performance: Produced the strongest regression results among all tested models. The model achieved an R^2 score of **0.920** on EGX30 and **0.940** on Gold, with a mean R^2 of approximately **0.930**. RMSE values remained low at **0.00418** and **0.00573**, respectively.

5. Random Forest

Purpose: Captures complex non-linear relationships using an ensemble of decision trees.

Configuration: Trained using $n_estimators=200$, $max_depth=8$, and $min_samples_leaf=5$ on the full feature set.

Performance: Achieved an R^2 score of **0.841** for EGX30 and **0.535** for Gold, with a mean R^2 of approximately **0.688**. The model showed stronger performance on EGX30 while maintaining reasonable generalization capability.

6. SVR (Support Vector Regressor)

Purpose: Models smooth non-linear return relationships using an epsilon-insensitive regression approach.

Configuration: Implemented with an RBF kernel using $C=1.0$, $epsilon=0.01$, and automatic gamma scaling.

Output: Achieved an R^2 score of **0.634** on EGX30 and **0.099** on Gold, with an average R^2 of approximately **0.367**. Although it produced the weakest standalone performance, the model remained useful for capturing alternative non-linear patterns and improving disagreement-based confidence estimation within the ensemble framework.

d. Visualization

Power BI Dashboard - Interactive Investment Analysis

An interactive Power BI dashboard was developed to analyze and compare the performance, volatility, and technical behavior of Egyptian market assets. The dashboard consists of three main pages.

i. Page 1: Market Overview

The first page provides a high-level market summary using the cleaned dataset before feature engineering. It includes KPI cards such as Total Return %, Current Price, and Sharpe Ratio.

- **Comparative Market Volatility:** compares daily percentage price changes across assets over time.
- **Asset Performance (Rebased to 100):** normalizes asset prices to compare growth trends regardless of original price levels.

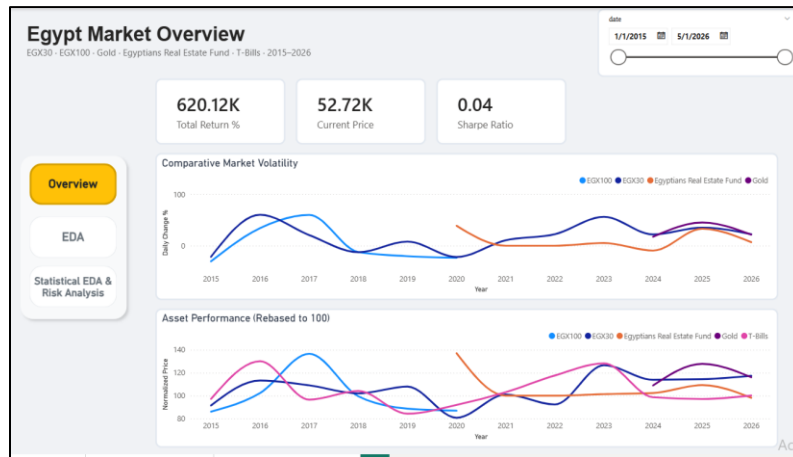


Figure 9: Dashboard Page1: Market Overview

ii. Page2: Exploratory Data Analysis (EDA) :

The second page focuses on individual asset analysis using the feature-engineered dataset. Users can select a single asset, and all visuals update accordingly.

The page includes:

- **Asset Price Trend** to visualize long-term price movement.
- **RSI** to indicate momentum and overbought/oversold conditions.
- **MACD Histogram** shows trend strength and momentum changes.



Figure 10: Dashboard Page2: Exploratory Data Analysis

iii. Page 3: Statistical EDA & Risk Analysis

The final page presents statistical and risk-based analysis across assets.

- **Risk vs. Return Scatter Plot** comparing volatility against total return.
- **Distribution of Daily Returns Histogram** showing return frequency and variability.
- **Technical Indicator Summary Table** comparing average RSI, MACD Histogram, Bollinger Band metrics, and distance from the 5-day SMA using conditional formatting.

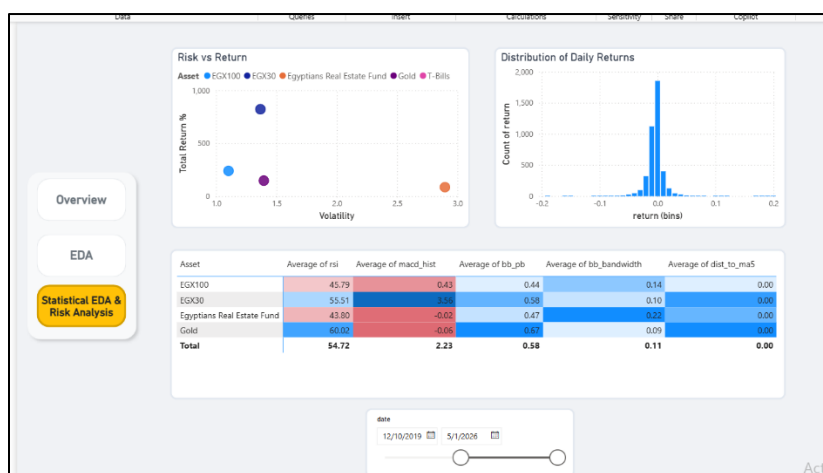


Figure 11: Dashboard Page3, Statistical EDA & Risk Analysis

Results

EDA Key findings

The exploratory analysis revealed noticeable differences in liquidity, volatility, and return behavior across assets. EGX30 and Gold showed higher volatility and stronger price movement compared to more stable assets. Correlation analysis also indicated varying relationships between asset classes, supporting the use of diversification in portfolio construction.

Additionally, some assets exhibited periods of stale or limited movement, reflecting lower market activity compared to more liquid instruments.

Regression Results (Strategic Forecasting)

The regression models were evaluated using R^2 , RMSE, and MAE metrics through walk-forward validation.

Linear Regression achieved the strongest overall performance, with an average R^2 score of approximately **0.93** across EGX30 and Gold, indicating that the engineered features captured a strong linear relationship with future returns.

Random Forest achieved moderate predictive performance with a mean R^2 of approximately **0.69**, while SVR produced the lowest average R^2 score of approximately **0.37**.

These results suggest that simpler linear relationships explained a significant portion of market behavior, while non-linear models captured additional but less consistent patterns.

Classification Results (Tactical Signals)

Classification models were evaluated using Accuracy, Precision, Recall, and weighted F1-score metrics.

The SVM classifier achieved the strongest tactical classification performance with approximately **37.8%** accuracy and **37.5%** weighted F1-score, slightly outperforming Logistic Regression, which achieved approximately **34.9%** accuracy and weighted F1-score.

Although classification accuracy remained close to the theoretical baseline of a three-class problem, the models still provided useful probabilistic signals and confidence estimates for tactical portfolio adjustments.

Montecarlo Key findings

Figure 1 illustrates the efficient frontier generated using the Monte Carlo simulation approach, highlighting the optimal investment portfolio in the Egyptian market based on the trade-off between risk and expected return. The figure demonstrates how different portfolio allocations perform under varying levels of volatility and return, with the optimal portfolio selected according to the highest risk-adjusted performance.

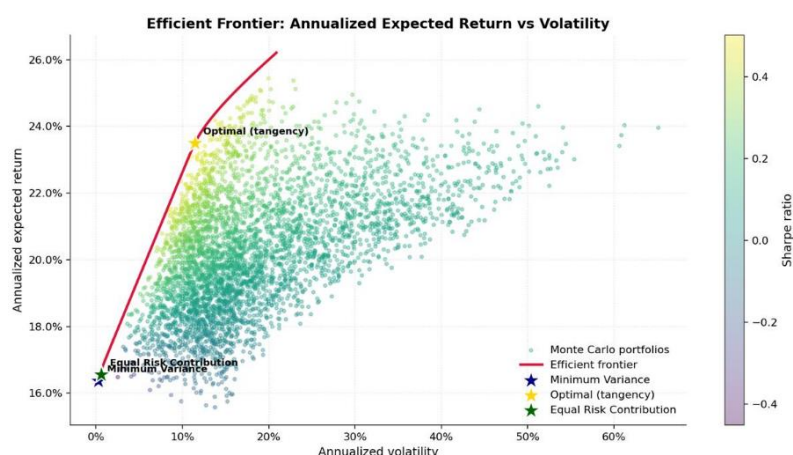


Figure 2 presents the asset weights corresponding to the optimal portfolio identified from the efficient frontier analysis. These weights represent the proportion of capital allocated to each asset in order to achieve the targeted balance between return maximization and risk minimization.

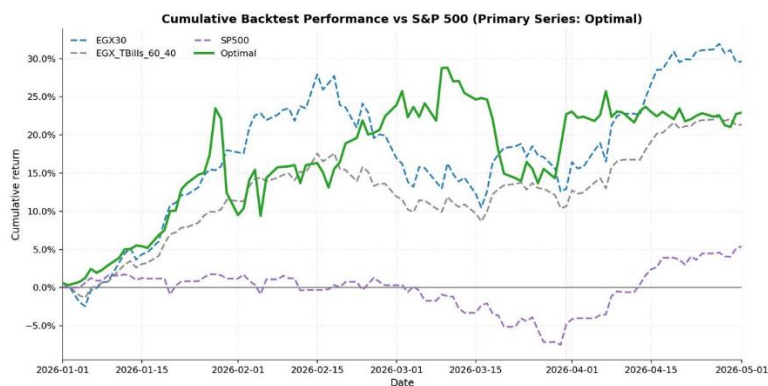


Figure 3 shows the backtesting results of the proposed portfolio compared with several benchmark portfolios and indices, including the EGX30, a 60/40 Treasury Bills allocation strategy, and the S&P 500. This comparison evaluates the historical performance of the selected portfolio and demonstrates its effectiveness relative to traditional investment strategies and market benchmarks.

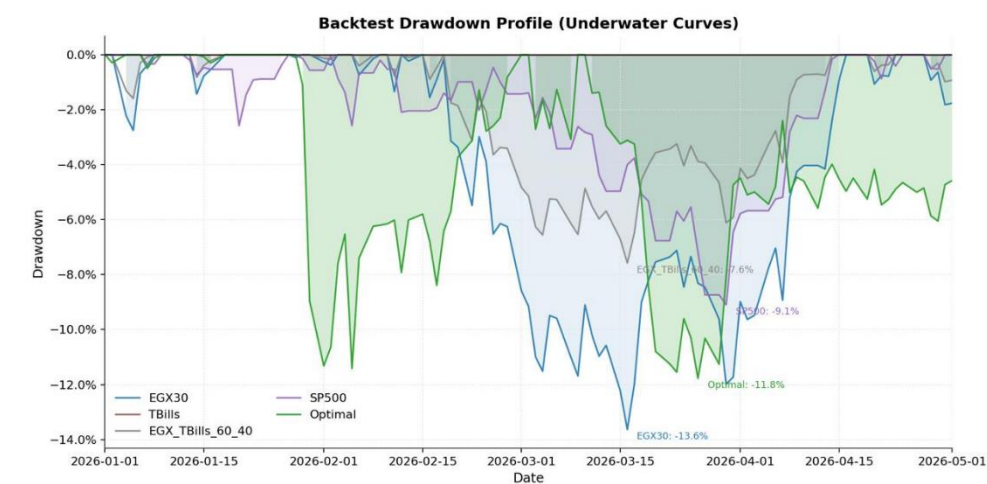
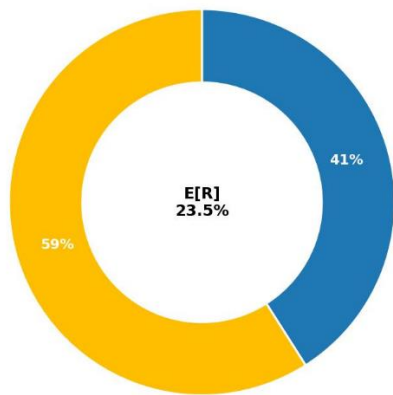


Figure 4 illustrates the drawdown analysis for the proposed portfolio and the benchmark portfolios. The figure highlights the magnitude of portfolio declines during adverse market conditions, providing insight into the downside risk and resilience of each investment strategy.

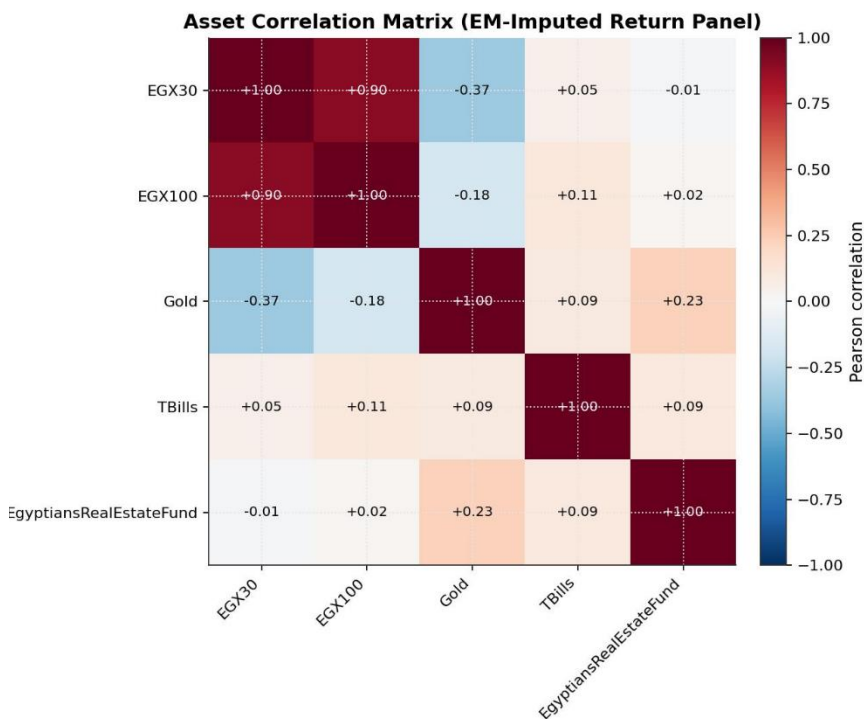
Strategic Asset Allocation (Optimal Portfolio)

Optimal Portfolio
Expected Volatility=11.4% Sharpe Ratio=+0.51



EGX30 EGX100 Gold TBills EgyptiansRealEstateFund

Moreover, the correlation heatmap presented in Figure 5 visualizes the correlation coefficients among all selected features and assets. This analysis helps identify the strength and direction of relationships between variables, offering a clearer understanding of diversification effects and interdependencies within the portfolio.



Conclusion

This study demonstrated how machine learning and data-driven portfolio analysis can support investment decision-making in the Egyptian market. Exploratory analysis highlighted differences in volatility, liquidity, and correlation across assets, emphasizing the value of diversification through assets such as Gold and T-Bills. Among the tested models, Linear Regression achieved the strongest regression performance, while XGBoost and SVM provided the most effective classification results for tactical market signals. Walk-forward backtesting further showed that combining predictive models with portfolio optimization improved risk-adjusted portfolio stability across different market conditions. Despite challenges related to market volatility, currency effects, and policy-driven regime shifts, the proposed framework provides a practical foundation for data-driven portfolio management and future expansion using real-time data and more advanced forecasting models.

Acknowledgement

Sincere appreciation is extended to Dr. Fatma Elshahaby and Dr. Malak Khaled for their valuable guidance, constructive feedback, and continuous support throughout the preparation of this technical report. Their expertise and encouragement were greatly appreciated and contributed significantly to the successful completion of this work.

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Appendix A: AI Audit

This appendix documents all instances where AI tools (e.g., ChatGPT, Claude) were used during this project, in compliance with CSE271 AI usage policy.

Instance 1:

- Tool used: [ChatGPT]
- Purpose: [Road map to learn using Power BI]
- Prompt used: [Paste your exact prompt — a screenshot must accompany this]
- AI output: [Summarize or paste the raw output — screenshot required]
- How we refined it: [e.g., "We verified all citations manually, rewrote the paragraph structure and added the CBE 2024 reference which the AI omitted."]

Instance 2:

- Tool used: [Gemeni]
- Purpose: [Getting insights on implementing Montecarlo, based on results from classical models]

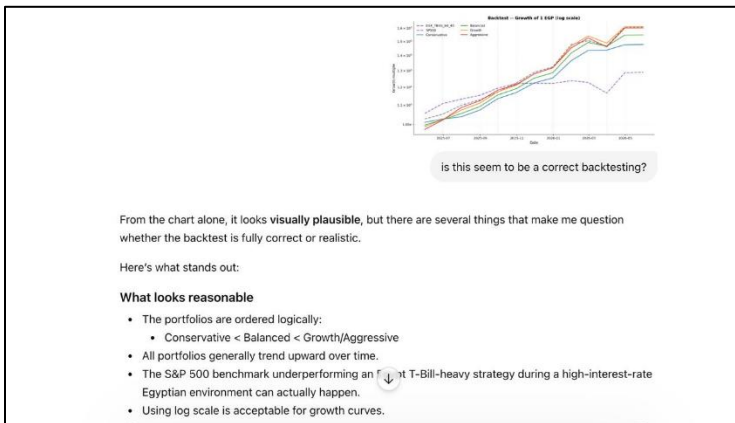
Instance 3:

- Tool used: [chat GPT]
- Purpose: [APA style format]
- Parmet [Put those references in APA style]

Instance 4:

- Tool used: [Claude]
- Purpose: [Fixing the formation of data sets]
- Parmet [passed on the formation of those first 20 rows, What kind of fixes show be applied to this data]

Note: A screenshot of most prompts is included below



- ◆ Evaluating a portfolio depends entirely on your **risk tolerance**, **investment horizon**, and **liquidity needs**. However, looking at this specific mix of Egyptian equities, fixed income, and gold, we can break down the construction.

Here is an analysis of your proposed portfolio:

1. Asset Breakdown & Role

Each component serves a specific purpose in an Egyptian market context:

- **EGX 30**: High-cap, liquid stocks. This is your primary **growth engine**, though it is sensitive to EGP fluctuations and global emerging market trends.
- **EGX 100**: Provides **broadier market exposure**, including mid and small-caps. There is significant overlap with the EGX 30, but it adds a layer of volatility and higher potential returns from smaller companies.

Ask Gemini

but regression will be used only for long term investor portfolios

and the classification model should be used for daily traders portfolio...

- ◆ Technically, **yes**, you can use them that way, but there is a nuance in how "professional" asset management handles these two worlds.

If you separate them entirely, you are essentially building two different products: a **Wealth Management tool** (for long-term investors) and a **Proprietary Trading tool** (for daily traders).

Here is how that breakdown works and where the risks lie:

1. The Long-Term Portfolio (Regression + MPT)

Your Logic: Correct. Regression is best at identifying "Value."

Ask Gemini

so, finally what should I do now?

- ◆ Google Search

To move from theory to reality, you should follow a structured "Pipeline Approach." Since you are targeting the Egyptian market, your next steps must prioritize **data sourcing** and **backtesting**, as these are the biggest hurdles in emerging markets.

Here is your immediate 4-step roadmap:

1. Data Procurement (The "EGP" Focus)

You cannot build these models without a clean dataset. For the Egyptian market, you need more than just prices.

- **Action**: Collect daily and monthly data for the last 10 years for:

Ask Gemini

+ Tools

Fast



These 2 layers will be for traders and long term investors together? ▾

2. Build the "Layer 1" Prototype (Strategic)
Before you trade daily, you need to know what the...

◆ Yes, exactly. Think of it as a **"shared engine"** that serves two different users by using different "outputs" from the same system.

By having both layers, you aren't just building a "bot"; you are building a professional **Investment Framework**. Here is how the two users interact with your two layers:

1. For the Long-Term Investor (The "Strategic" User)

This user primarily cares about **Layer 1**. They want a portfolio that grows steadily and beats inflation over 1–5 years.

Ask Gemini

+ Tools Fast ▾

give me road map to start using power bi

◆ Show thinking ▾

Starting with Power BI is a fantastic career move. It is a powerful tool, but it can be overwhelming at first because it's actually three tools built into one: **Power Query** (for data prep), the **Data Model** (for structuring), and **DAX** (the formula language).

Here is a structured, step-by-step roadmap to take you from a complete beginner to a confident Power BI user.

Phase 1: The Fundamentals & Setup

Your goal: Get familiar with the software and understand the workflow.

Enter a prompt for Gemini

give me a guide to get the EGX 30 historcial data, you do not give me just outlines, I want every detail of getting it

Searched the web >

Here is a complete, detailed guide to getting EGX 30 historical data — covering every method available, step by step.

Complete Guide to Getting EGX 30 Historical Data

There are four main ways to get this data, ranging from clicking a button on a website to writing Python code. All are covered below in full detail.

Method 1: Investing.com (Best Free Mar 1 Source)

This is the most convenient option and requires no coding.

I tried to understand what is the meaning of T-bills from many sources, can you now give me a full explanation



A Treasury Bill (T-bill) is basically a way for the government to **borrow money from people for a short period of time**.

You give the government money today.

After a few months, the government gives you back **more money**.

That's the entire core idea.

The Simplest Real Example

Imagine the government says:

"Give me 98,000 EGP today, and after 1 year I'll return 100,000 EGP."

...

You agree.

- You pay: 98,000

How to read the heat map

Reading a heat map is basically learning how to translate:

colors → values

Once you know what the colors mean, the whole map becomes understandable.

Step 1 — Look at the Color Scale (Legend)

This is the MOST important thing.

Every heat map has a color bar/legend somewhere.

...

Example:

Get the color bar/legend from the map

Chan, E. P. (2013). *Algorithmic trading: Winning strategies and their rationale*. John Wiley & Sons.
Narang, R. K. (2013). *Inside the black box: A simple guide to quantitative and high-frequency trading* (2nd ed.). John Wiley & Sons.
Grinold, R. C., & Kahn, R. N. (1999). *Active portfolio management: A quantitative approach for producing superior returns and controlling risk* (2nd ed.). McGraw-Hill.
Dixon, M. F., Halperin, I., & Bilokon, P. (2020). *Machine learning in finance: From theory to practice*. Springer.
Put them in APA style



- Algorithmic Trading: Winning Strategies and Their Rationale
Chan, E. P. (2013). *Algorithmic trading: Winning strategies and their rationale*. John Wiley & Sons.
- Inside the Black Box: A Simple Guide to Quantitative and High-Frequency Trading
Narang, R. K. (2013). *Inside the black box: A simple guide to quantitative and high-frequency trading* (2nd ed.). John Wiley & Sons.
- Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling

list the most five trusted financial data websites

Here are the 5 most trusted financial data websites used by investors, analysts, traders, and institutions worldwide:

1. Bloomberg ↗

Probably the most respected financial data source in the world.

Why it's trusted:

- Institutional-grade data
- Extremely fast and accurate
- Used by:
 - banks
 - hedge funds